

Lecture Example 3 - Big-M with an Artificial Variable

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:54

Lecture slides 60-77. A $\geq$ constraint requires a surplus and an artificial variable penalised by $-M$ in the objective.

Model

Maximize $Z = 4X_1 + 3X_2$
subject to
[C1] $X_1 + 2X_2 \leq 4$
[C2] $2X_1 + X_2 \geq 6$
[C3] $X_2 \leq 1$
 $X_1, X_2 \geq 0$

RESULT

OPTIMAL

$Z^* = 16$

$X_1 = 4$

$X_2 = 0$

iterations: 2 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 16$

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X3$ is the starting basic variable of this row.
2. C2 (Type II, $\geq$): subtract surplus $X4 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis $\rightarrow$ an artificial variable is needed.
3. C3 (Type I, $\leq$): add slack $X5 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X5$ is the starting basic variable of this row.
4. C2: add artificial $X6 \geq 0$ with objective coefficient $-M$ (huge penalty) $\rightarrow$ it provides an immediately recognisable starting basic variable.

Maximize $Z = 4X1 + 3X2 - MX6$

subject to

$$X1 + 2X2 + X3 = 4$$

$$2X1 + X2 - X4 + X6 = 6$$

$$X2 + X5 = 1$$

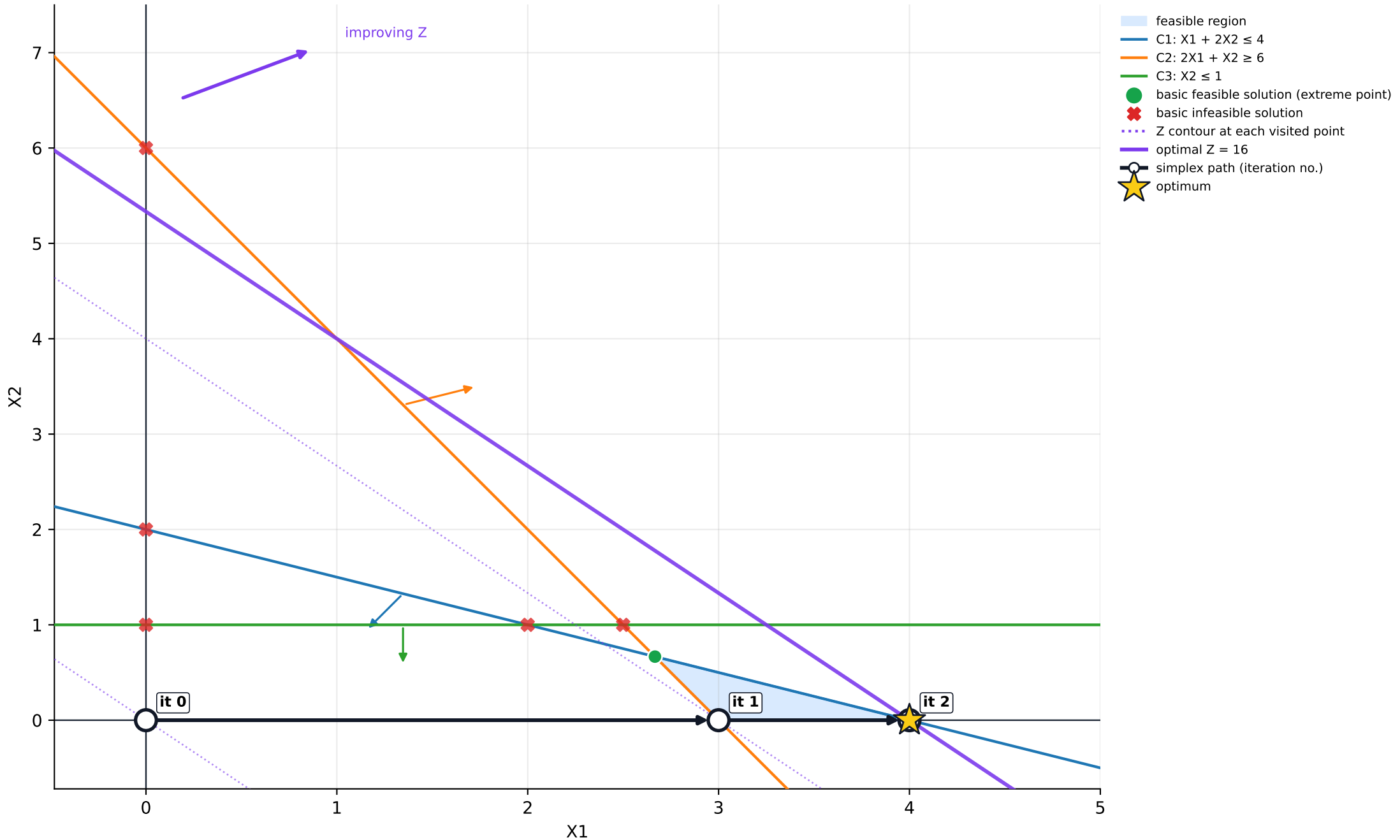
$$X1, X2, X3, X4, X5, X6 \geq 0$$

Standard form - coefficient matrix

	X1	X2	X3	X4	X5	X6		RHS
max Z	4	3	0	0	0	-M		
C1	1	2	1	0	0	0	=	4
C2	2	1	0	-1	0	1	=	6
C3	0	1	0	0	1	0	=	1

2. Geometric view: basic solutions, extreme points and the simplex path

Geometry of the simplex path - optimal



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	X5	Optimal?
X1, X2, X3	2.5	1	no	13	-0.5	0	0	
X1, X2, X4	2	1	no	11	0	-1	0	
X1, X2, X5	2.6667	0.6667	yes	12.6667	0	0	0.3333	
X1, X3, X5	3	0	yes	12	1	0	1	
X1, X4, X5	4	0	yes	16	0	2	1	★
X2, X3, X4	0	1	no	3	2	-5	0	
X2, X3, X5	0	6	no	18	-8	0	-5	
X2, X4, X5	0	2	no	6	0	-4	-1	
X3, X4, X5	0	0	no	0	4	-6	1	

Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	-4	-3	0	0	0	M	0
X3	1	0	1	2	1	0	0	0	4
X6	2	0	2	1	0	-1	0	1	6
X5	3	0	0	1	0	0	1	0	1

Initial tableau. Basic variables: $X3 = 4$, $X6 = 6$, $X5 = 1$. Non-basic (set to 0): $X1$, $X2$, $X4$. The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

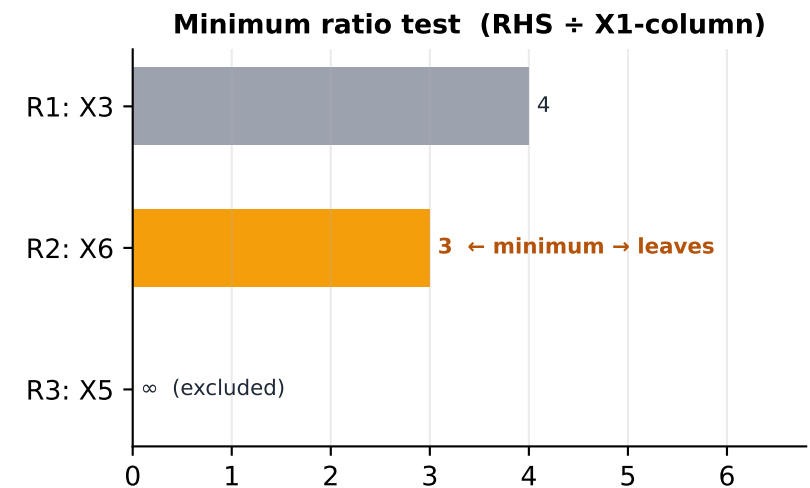
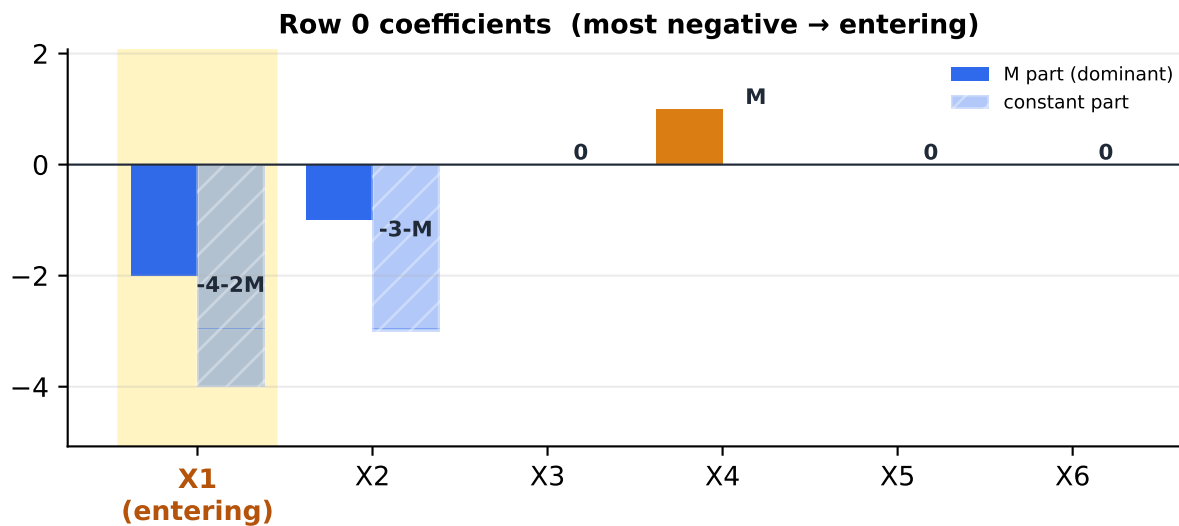
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	-4-2M	-3-M	0	M	0	0	-6M
X3	1	0	1	2	1	0	0	0	4
X6	2	0	2	1	0	-1	0	1	6
X5	3	0	0	1	0	0	1	0	1

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting $M \times$ their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract $M \times R2$ from $R0$ (make the coefficient of basic X6 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	-4-2M	-3-M	0	M	0	0	-6M	
X3	1	0	1	2	1	0	0	0	4	4
X6	2	0	2	1	0	-1	0	1	6	3
X5	3	0	0	1	0	0	1	0	1	∞



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (-4-2M).
- Minimum ratio test (RHS $\div$ positive entries of the X1-column): R1: 4, R2: 3, R3: ∞ .
- Leaving basic variable: X6 (row 2, smallest ratio). Pivot number = 2.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 2

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	-4-2M	-3-M	0	M	0	0	-6M	
X3	1	0	1	2	1	0	0	0	4	4
X6	2	0	1	0.5	0	-0.5	0	0.5	3	3
X5	3	0	0	1	0	0	1	0	1	∞

(b) X1 replaced X6 in the basis. Z = 12

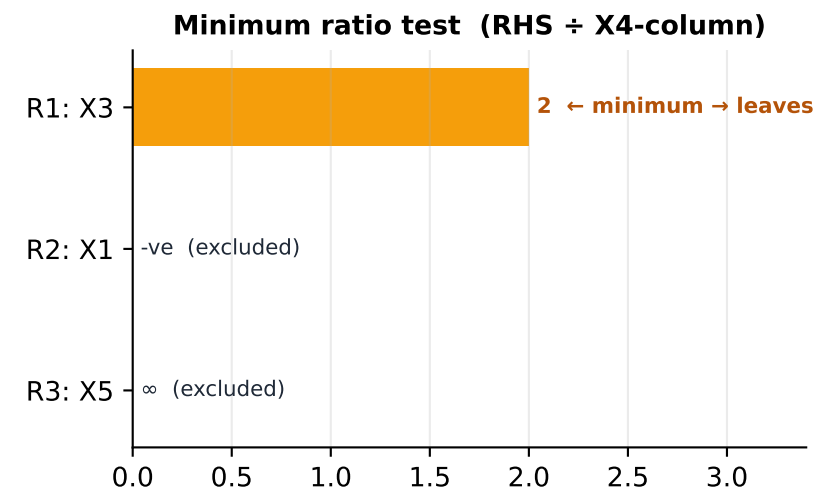
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	-1	0	-2	0	2+M	12
X3	1	0	0	1.5	1	0.5	0	-0.5	1
X1	2	0	1	0.5	0	-0.5	0	0.5	3
X5	3	0	0	1	0	0	1	0	1

Row operations:

- Add $(4+2M) \times R2$ to R0
- Subtract $1 \times R2$ from R1

Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	-1	0	-2	0	2+M	12	
X3	1	0	0	1.5	1	0.5	0	-0.5	1	2
X1	2	0	1	0.5	0	-0.5	0	0.5	3	-ve
X5	3	0	0	1	0	0	1	0	1	∞



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X4 - it has the largest (in absolute value) negative coefficient in row 0 (-2).
- Minimum ratio test (RHS ÷ positive entries of the X4-column): R1: 2, R2: -ve, R3: ∞ .
- Leaving basic variable: X3 (row 1, smallest ratio). Pivot number = 0.5.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 0.5

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	-1	0	-2	0	2+M	12	
X3	1	0	0	3	2	1	0	-1	2	2
X1	2	0	1	0.5	0	-0.5	0	0.5	3	-ve
X5	3	0	0	1	0	0	1	0	1	∞

(b) X4 replaced X3 in the basis. Z = 16

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	5	4	0	0	M	16
X4	1	0	0	3	2	1	0	-1	2
X1	2	0	1	2	1	0	0	0	4
X5	3	0	0	1	0	0	1	0	1

Row operations:

- Add $2 \times R1$ to R0
- Add $0.5 \times R1$ to R2

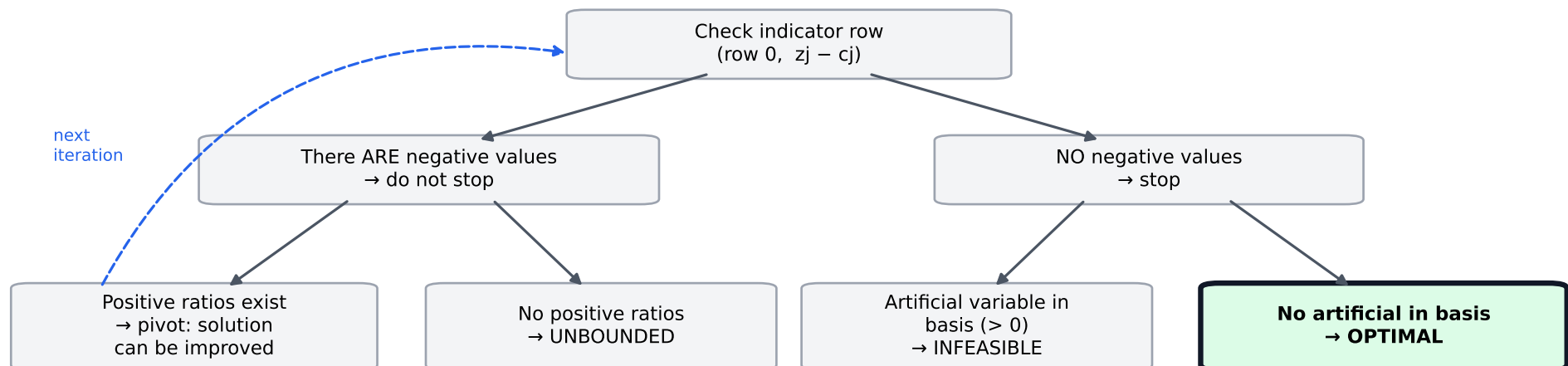
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	5	4	0	0	M	16
X4	1	0	0	3	2	1	0	-1	2
X1	2	0	1	2	1	0	0	0	4
X5	3	0	0	1	0	0	1	0	1

Final check.

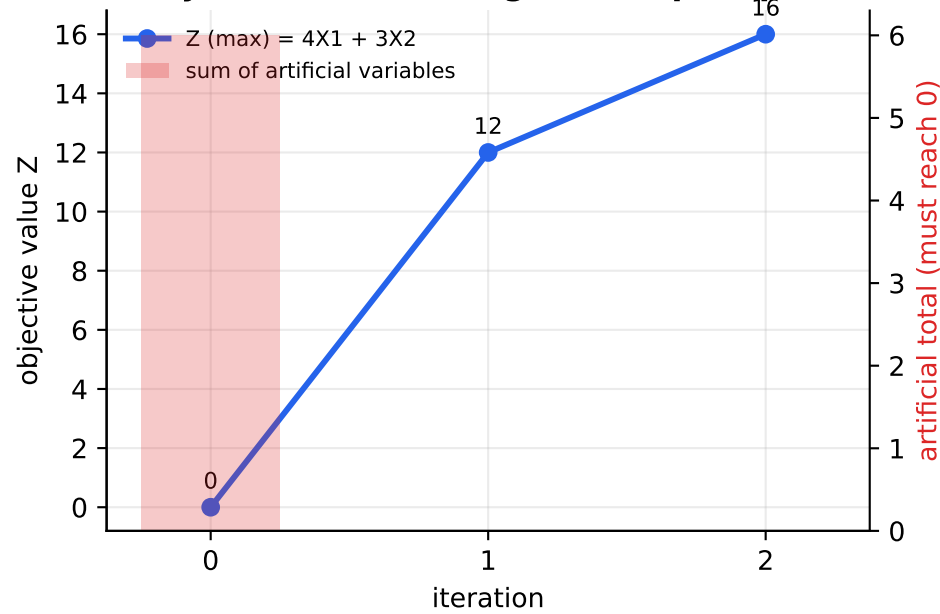
- Optimal: row 0 has no negative coefficients. $Z^* = 16$

Simplex decision diagram - outcome: OPTIMAL

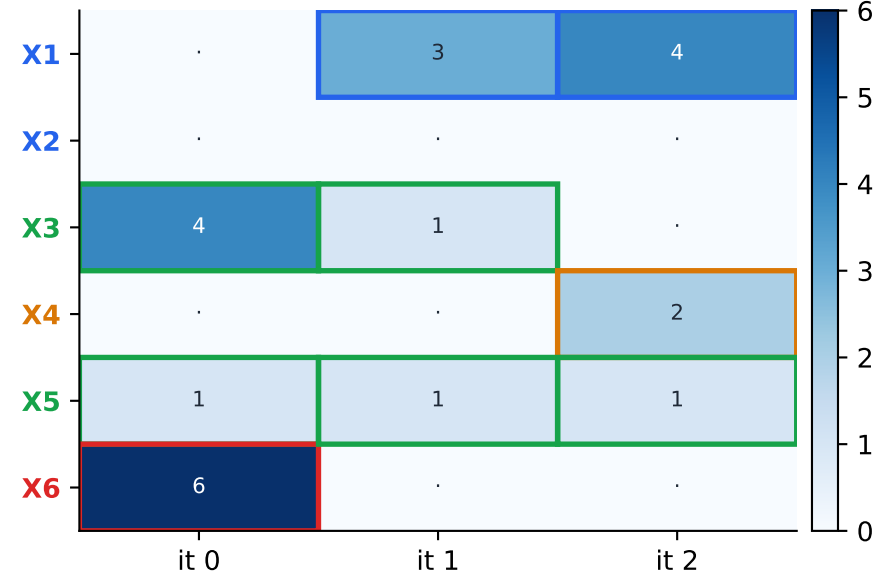


Lecture Example 3 - Big-M with an Artificial Variable - OPTIMAL, $Z^* = 16$

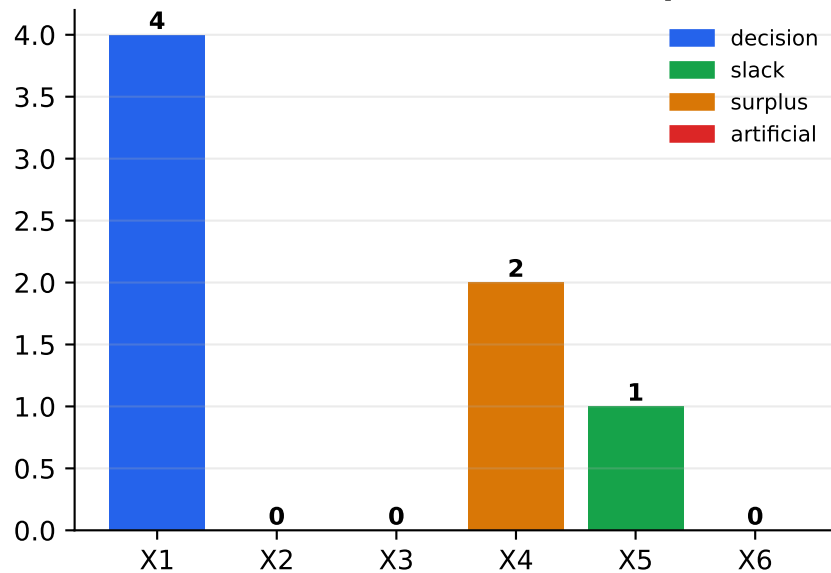
Objective value along the simplex path



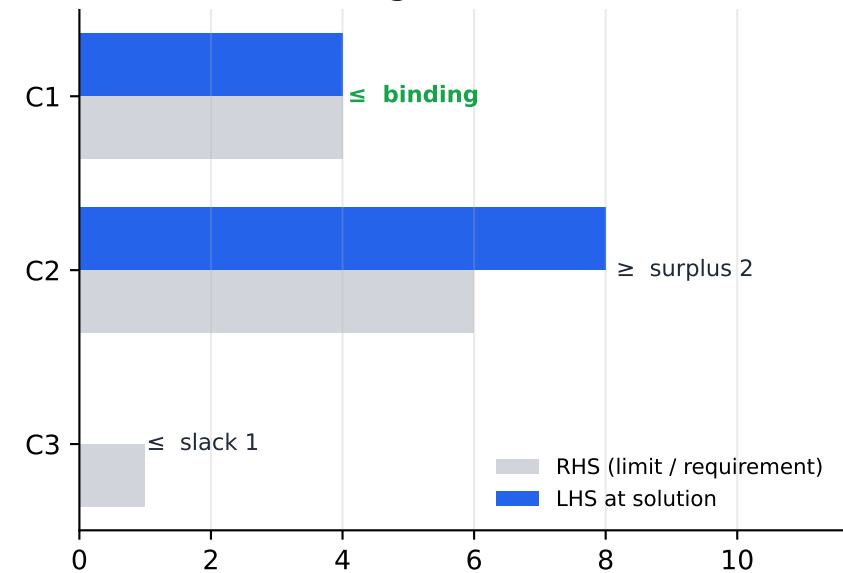
Basis evolution (outlined = basic variable)



Final values of all variables (optimal)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	16	16
X1	4	4
X2	0	0

✓ **Results agree**